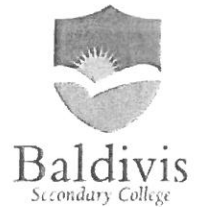


Year 8 Mathematics Semester 2 Exam



Resources Allowed: One A4 page of notes (2-sided) and calculator

Time: 60 mins Name: T / 60 marks

Full working out must be shown to get full marks. Attempt all the questions.

Part A

21 marks

1. For each word below, write the letter of the matching definition.

[3 marks]

C Probability Experiment
A Outcome
B Sample Space
E Event
F Complement
D Trial

A: A possible result.
B: All possible results.
C: An activity that involves chance.
D: Doing the activity.
E: A particular result.
F: Not a particular result.

2. For each of the following equations find the value of x .

[6 marks]

a) $x + 7 = 12$

$$\begin{array}{r} -7 \quad -7 \\ x + 7 = 12 \\ \hline x = 5 \end{array}$$

b) $2x + 3 = 17$

$$\begin{array}{r} -3 \quad -3 \\ 2x + 3 = 17 \\ \hline 2x = 14 \\ \div 2 \quad \div 2 \\ x = 7 \end{array}$$

c) $3(x + 4) = 18$

$$\begin{array}{r} \div 3 \quad \div 3 \\ 3(x + 4) = 18 \\ \hline x + 4 = 6 \\ -4 \quad -4 \\ x = 2 \end{array}$$

d) $2x + 6 = 4(x - 2)$

$$\begin{array}{r} 2x + 6 = 4x - 8 \\ 6 + 8 = 4x - 2x \\ 14 = 2x \\ x = 7 \end{array}$$

3. The letters of the word STATISTICS are written on 10 cards. The cards are placed in a hat and one card is chosen at random.

[4 marks]

- a) State the sample space

SS: {S, T, A, T, I, S, T, I, C, S}

- c) Find the probability of choosing a vowel

$$\frac{3}{10}$$

- b) Find the probability of choosing the letter T

$$\frac{3}{10}$$

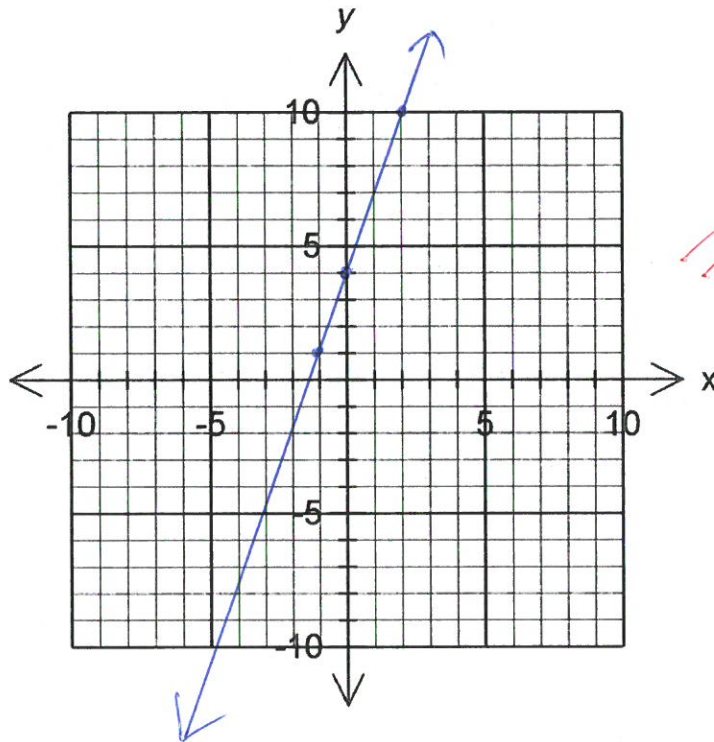
- d) What is the probability that the letter chosen will be in the word STATS?

$$\frac{7}{10}$$

4. Fill in the table and then plot as a linear graph

[4 marks]

x	-2	-1	0	1	2
y	-2	1	4	7	10



5. [2 marks]

A taxi departs at 18:32 from Baldivis and arrives at 20:13 at Armadale. How long did the taxi take for the trip?

18:32 → 19:32 1hr
 19:32 → 20:00 28mins
 20:00 → 20:13 13mins

 1hr 41mins

6. [2 marks]

Ryles in Broome is expecting a phone call from Ella in Sydney. Broome is 3 hours behind Sydney time.

a) If Ryles calls at 8:45am, what time is it for Ella?

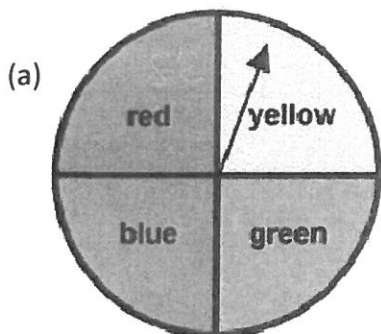
11:45pm

b) If Ella calls at 1:30pm, what time is it for Ryles?

10:30am

7. [4 marks]

The following fair spinners are spun. Write the probability, as a fraction, decimal and percentage, of landing on red. What is the probability, as a decimal, of NOT landing on red?

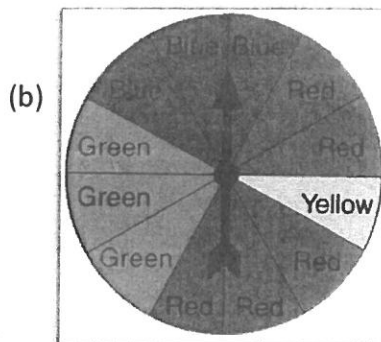


Fraction: $\frac{1}{4}$

Decimal: 0.25

Percentage: 25%

Pr (R'): $\frac{3}{4}$



F $\frac{5}{12}$

D 0.42

% 42%

$\frac{7}{12}$

8. [6 marks]

Consider the following set of data

53, 37, 53, 42, 10, 49, 42, 53, 45, 47
10, 37, 42, 42, 45, 47, 49, 53, 53, 53

a) Calculate;

I. Mean

$$431 \div 10 = 43.1$$

II. Median

46

III. Mode

53

IV. Range

43

b) Which score in the set of data is an outlier?

10

c) If the outlier is removed, which of the above measures will it affect the most? (Mean, Median, Mode or Range)

Mean / Range

9. [6 marks]

The two-way table below shows the results of a poll conducted of a group of students who own mobile phones to see who pays their own bills.

	Boys	Girls	Total
Pay own bill	7	16	23
Do not pay own bill	5	4	9
Total	12	20	32

- a) Complete the above two-way table
b) How many people participated in this poll?

32

- c) How many of the students surveyed did not pay their own bill?

9

- d) Find the **probability** that a randomly selected student:

- i. Is a girl who pays her own bill

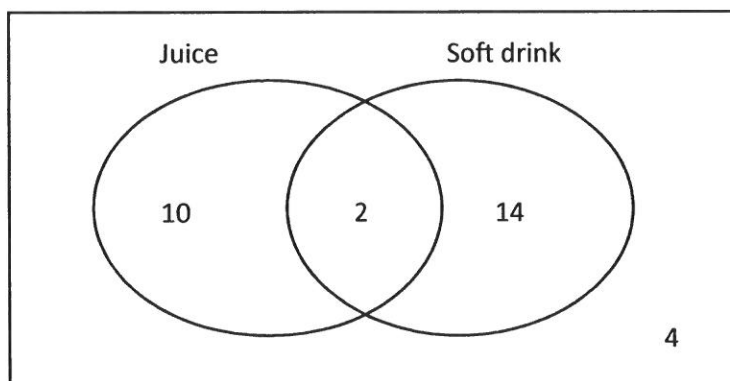
$$\frac{16}{32} = \frac{1}{2}$$

- ii. Does not pay their own bill

$$\frac{9}{32}$$

10. [6 marks]

The following Venn Diagram shows the number of people who like juice and/or soft drinks



- a) What is the total number of people who like juice?

12

- b) What is the number of people who only like soft drink?

14

- c) What is the **probability**:

- i. That a randomly selected person likes neither juice nor soft drink?

$$\frac{4}{30} = \frac{2}{15}$$

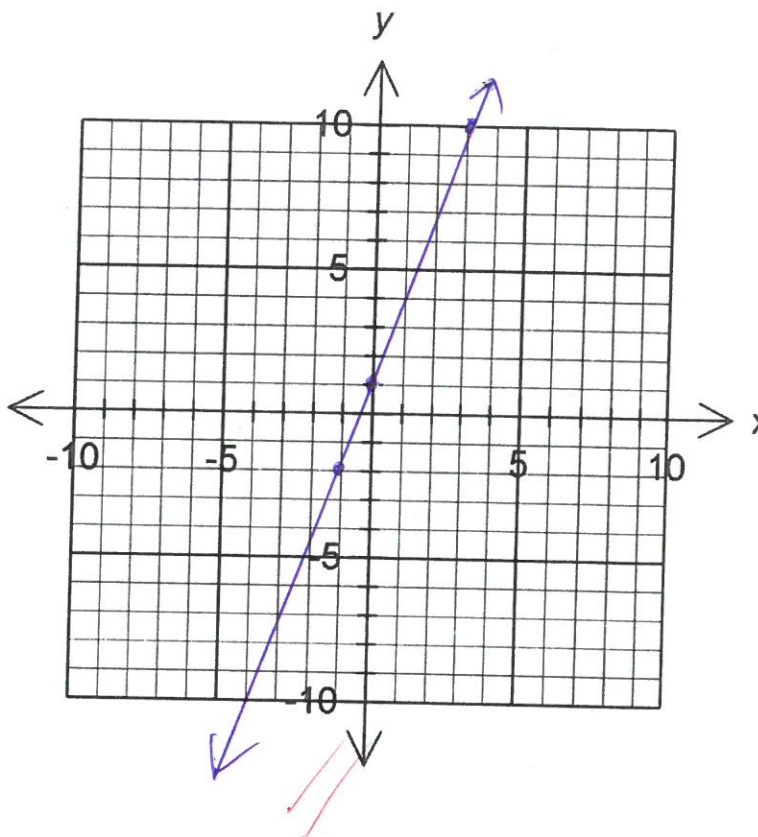
- ii. That a randomly selected person likes either juice or soft drink or both?

$$\frac{26}{30} = \frac{13}{15}$$

11. [4 marks]

Use the rule $y = 3x + 1$ to fill in the table below and then graph it

x	-1	0	1	2	3
y	-2	1	4	7	10



12. [4 marks]

Find the gradient and y-intercepts for these lines

(a) Gradient = 3

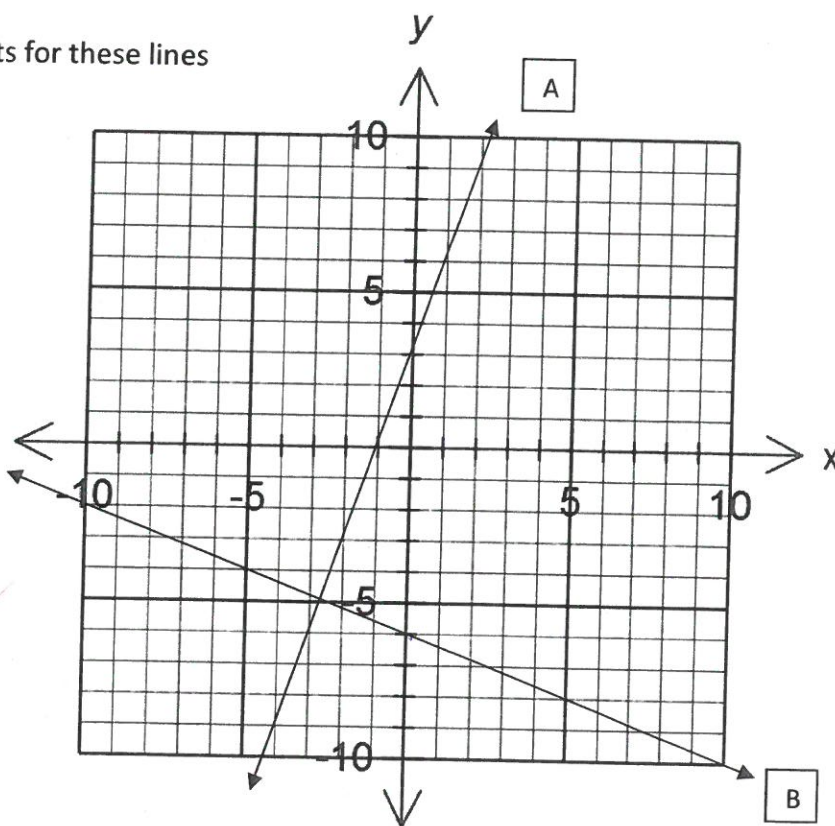
Y-intercept = 3

Rule: $y = 3x + 3$

(b) Gradient = $-\frac{2}{5}$

Y-intercept = -6

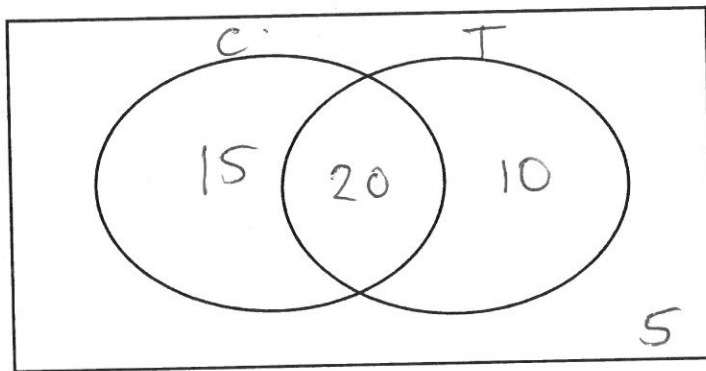
Rule: $y = -\frac{2}{5}x - 6$



13. [9 marks]

A survey is conducted of 50 people, asking who likes coffee and who likes tea. It was found that 20 people liked both, 35 people liked coffee, and 30 people liked tea.

a) Represent this using a Venn diagram. (3)



b) How many people liked neither tea nor coffee? (1)

5

c) What is the **probability** of people liking coffee AND tea? (1)

$$\frac{20}{50} = \frac{2}{5}$$

d) What is the **probability** of people liking coffee OR tea? (1)

$$\frac{45}{50} = \frac{9}{10}$$

e) Present the survey findings in a two-way table. (3)

	C	C'	Total
T	20	10	30
T'	15	5	20
Total	35	15	50